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4.32 a. The pmf of Y , for $y = 0, 1, \dots$, is

$$\begin{aligned} f_Y(y) &= \int_0^\infty f_Y(y|\lambda) f_\Lambda(\lambda) d\lambda = \int_0^\infty \frac{\lambda^y e^{-\lambda}}{y!} \frac{1}{\Gamma(\alpha)\beta^\alpha} \lambda^{\alpha-1} e^{-\lambda/\beta} d\lambda \\ &= \frac{1}{y! \Gamma(\alpha) \beta^\alpha} \int_0^\infty \lambda^{(y+\alpha)-1} \exp\left\{\frac{-\lambda}{\left(\frac{\beta}{1+\beta}\right)}\right\} d\lambda \\ &= \frac{1}{y! \Gamma(\alpha) \beta^\alpha} \Gamma(y+\alpha) \left(\frac{\beta}{1+\beta}\right)^{y+\alpha}. \end{aligned}$$

If α is a positive integer,

$$f_Y(y) = \binom{y+\alpha-1}{y} \left(\frac{\beta}{1+\beta}\right)^y \left(\frac{1}{1+\beta}\right)^\alpha,$$

the negative binomial($\alpha, 1/(1+\beta)$) pmf. Then

$$\begin{aligned} EY &= E(E(Y|\Lambda)) = E\Lambda = \alpha\beta \\ \text{Var}Y &= \text{Var}(E(Y|\Lambda)) + E(\text{Var}(Y|\Lambda)) = \text{Var}\Lambda + E\Lambda = \alpha\beta^2 + \alpha\beta = \alpha\beta(\beta+1). \end{aligned}$$

b. For $y = 0, 1, \dots$, we have

$$\begin{aligned}
 P(Y = y|\lambda) &= \sum_{n=y}^{\infty} P(Y = y|N = n, \lambda)P(N = n|\lambda) \\
 &= \sum_{n=y}^{\infty} \binom{n}{y} p^y (1-p)^{n-y} \frac{e^{-\lambda} \lambda^n}{n!} \\
 &= \sum_{n=y}^{\infty} \frac{1}{y!(n-y)!} \left(\frac{p}{1-p}\right)^y [(1-p)\lambda]^n e^{-\lambda} \\
 &= e^{-\lambda} \sum_{m=0}^{\infty} \frac{1}{y!m!} \left(\frac{p}{1-p}\right)^y [(1-p)\lambda]^{m+y} \quad (\text{let } m = n - y) \\
 &= \frac{e^{-\lambda}}{y!} \left(\frac{p}{1-p}\right)^y [(1-p)\lambda]^y \left[\sum_{m=0}^{\infty} \frac{[(1-p)\lambda]^m}{m!} \right] \\
 &= e^{-\lambda} (p\lambda)^y e^{(1-p)\lambda} \\
 &= \frac{(p\lambda)^y e^{-p\lambda}}{y!},
 \end{aligned}$$

the Poisson($p\lambda$) pmf. Thus $Y|\Lambda \sim \text{Poisson}(p\lambda)$. Now calculations like those in a) yield the pmf of Y , for $y = 0, 1, \dots$, is

$$f_Y(y) = \frac{1}{\Gamma(\alpha)y!(p\beta)^\alpha} \Gamma(y+\alpha) \left(\frac{p\beta}{1+p\beta}\right)^{y+\alpha}.$$

Again, if α is a positive integer, $Y \sim \text{negative binomial}(\alpha, 1/(1+p\beta))$.

2.

4.35 a. $\text{Var}(X) = \text{E}(\text{Var}(X|P)) + \text{Var}(\text{E}(X|P))$. Therefore,

$$\begin{aligned}
 \text{Var}(X) &= \text{E}[nP(1-P)] + \text{Var}(nP) \\
 &= n \frac{\alpha\beta}{(\alpha+\beta)(\alpha+\beta+1)} + n^2 \text{Var}P \\
 &= n \frac{\alpha\beta(\alpha+\beta+1-1)}{(\alpha+\beta^2)(\alpha+\beta+1)} + n^2 \text{Var}P \\
 &= \frac{n\alpha\beta(\alpha+\beta+1)}{(\alpha+\beta^2)(\alpha+\beta+1)} - \frac{n\alpha\beta}{(\alpha+\beta^2)(\alpha+\beta+1)} + n^2 \text{Var}P \\
 &= n \frac{\alpha}{\alpha+\beta} \frac{\beta}{\alpha+\beta} - n \text{Var}P + n^2 \text{Var}P \\
 &= n \text{E}P(1-\text{E}P) + n(n-1) \text{Var}P.
 \end{aligned}$$

b. $\text{Var}(Y) = \text{E}(\text{Var}(Y|\Lambda)) + \text{Var}(\text{E}(Y|\Lambda)) = \text{E}\Lambda + \text{Var}(\Lambda) = \mu + \frac{1}{\alpha}\mu^2$ since $\text{E}\Lambda = \mu = \alpha\beta$ and $\text{Var}(\Lambda) = \alpha\beta^2 = \frac{(\alpha\beta)^2}{\alpha} = \frac{\mu^2}{\alpha}$. The “extra-Poisson” variation is $\frac{1}{\alpha}\mu^2$.

3.

For any two random variables X and Y with finite variances, prove that

(a) $E(XY) = E(XE(Y|X))$.

Let $f(x, y)$ be the joint density for X and Y .

$$\begin{aligned} E(XY) &= \int_x \int_y xy f(x, y) dy dx \\ &= \int_x \int_y xy f(y|x) f(x) dy dx \\ &= \int_x x f(x) \left(\int_y y f(y|x) dy \right) dx \\ &= \int_x x f(x) E(Y|X) dx \\ &= E(XE(Y|X)) \end{aligned}$$

(b) $Cov(X, Y) = Cov(X, E(Y|X))$.

$$\begin{aligned} Cov(X, Y) &= E(XY) - E(X)E(Y) \\ &= E[XE(Y|X)] - E(X)E[E(Y|X)] \\ &= Cov(X, E(Y|X)) \end{aligned}$$

(c) X and $Y - E(Y|X)$ are uncorrelated.

$$\begin{aligned} Cov(X, Y - E(Y|X)) &= E[X(Y - E(Y|X))] - E(X)E[Y - E(Y|X)] \\ &= E[XE(Y|X)] - E(X)E(Y) + E(X)E[E(Y|X)] \\ &= E(XY) - E[XE(Y|X)] - E(X)E(Y) + E(X)E(Y) \quad \text{by Theorem 4.4.3} \\ &= E(XY) - E(XY) \quad \text{from part a} \\ &= 0 \end{aligned}$$

Since $Cov(X, Y - E(Y|X)) = 0$, the correlation will also be 0.

4.

a. $E(x) = E(E(x|Y)) = nE(Y) = \frac{na}{a+b}$

(b) $E(x) = E\left(\frac{1}{Y}\right) = \frac{b}{a-1}$ (using knowledge of Inverse Gamma)

5.

$$Var(x) = E(Var(x|Y)) + Var(E(x|Y))$$

$$= E(nY(1-Y)) + Var(nY)$$

$$= nE(Y - Y^2) + n^2 Var(Y)$$

$$= \frac{na}{a+b} - \frac{na}{(a+b)^2(a+b+1)} - \frac{na^2}{(a+b)^2} + \frac{n^2 ab}{(a+b)^2(a+b+1)} \quad \downarrow$$

) cont. $= \frac{1}{(a+b)^2(a+b+1)} [na(a+b)(a+b+1) - nab - na^2(a+b+1) + n^2 ab]$

$$= \frac{na}{(a+b)^2(a+b+1)} [a^2 + ab + a + tab + b^2 + b - b - a^2 - ab - a + nb]$$

$$= \frac{na}{(a+b)^2(a+b+1)} [ab + b^2 + nb]$$

$$= \frac{nab(a+b+n)}{(a+b)^2(a+b+1)}$$

6.

$$\begin{aligned}
E(e^{tx}) &= E(E(e^{tx} | Y)) \\
&= E(e^{Y(e^{t-1})}) \\
&= e^{1(e^{t-1})}(.3) + e^{2(e^{t-1})}(.7) \\
&= .3e^{e^{t-1}} + .7e^{2e^{t-1}} \\
&= e^{e^{t-1}}(.3 + .7e^{e^{t-1}})
\end{aligned}$$

4.41 Let a be a constant. $\text{Cov}(a, X) = E(aX) - EaEX = aEX - aEX = 0$.

4.42

$$\rho_{XY,Y} = \frac{\text{Cov}(XY, Y)}{\sigma_{XY}\sigma_Y} = \frac{E(XY^2) - \mu_{XY}\mu_Y}{\sigma_{XY}\sigma_Y} = \frac{EXEY^2 - \mu_X\mu_Y\mu_Y}{\sigma_{XY}\sigma_Y},$$

where the last step follows from the independence of X and Y . Now compute

$$\begin{aligned}
\sigma_{XY}^2 &= E(XY)^2 - [E(XY)]^2 = EX^2EY^2 - (EX)^2(EY)^2 \\
&= (\sigma_X^2 + \mu_X^2)(\sigma_Y^2 + \mu_Y^2) - \mu_X^2\mu_Y^2 = \sigma_X^2\sigma_Y^2 + \sigma_X^2\mu_Y^2 + \sigma_Y^2\mu_X^2.
\end{aligned}$$

Therefore,

$$\rho_{XY,Y} = \frac{\mu_X(\sigma_Y^2 + \mu_Y^2) - \mu_X\mu_Y^2}{(\sigma_X^2\sigma_Y^2 + \sigma_X^2\mu_Y^2 + \sigma_Y^2\mu_X^2)^{1/2}\sigma_Y} = \frac{\mu_X\sigma_Y}{(\mu_X^2\sigma_Y^2 + \mu_Y^2\sigma_X^2 + \sigma_X^2\sigma_Y^2)^{1/2}}.$$

4.43

$$\begin{aligned}
\text{Cov}(X_1 + X_2, X_2 + X_3) &= E(X_1 + X_2)(X_2 + X_3) - E(X_1 + X_2)E(X_2 + X_3) \\
&= (4\mu^2 + \sigma^2) - 4\mu^2 = \sigma^2 \\
\text{Cov}(X_1 + X_2, X_1 - X_2) &= E(X_1 + X_2)(X_1 - X_2) = EX_1^2 - X_2^2 = 0.
\end{aligned}$$

(4.65) Consider $q = (g(x) - g(y))(h(x) - h(y))$ for $X, Y \stackrel{\text{iid}}{\sim} F(\cdot)$. This will have a helpful $E(q)$

(a) If g is non-decreasing & h is non-decreasing, then
for $X > y$, $q \geq 0$
 $X < y$, $q \leq 0$. So, $q \leq 0$.
 $X = y$, $q = 0 \leq 0$

(b) If g & h are both non-decreasing or non-increasing, then
when $X > y$, $q \geq 0$
 $X < y$, $q \geq 0$
 $X = y$, $q = 0 \geq 0$.

Now, note

$$\begin{aligned} E(q) &= E[g(x)h(x)] + E[g(y)h(y)] - \\ &\quad E[g(y)h(x)] - E[g(x)h(y)] \\ &= 2E(g(x)h(x)) - E[g(y)]E[h(x)] - E[g(x)]E[h(y)] \end{aligned}$$

Since X, Y are iid.

$$= 2E(g(x)h(x)) - 2E(g(x))E(h(x)) \text{ since } X, Y \text{ are iid.}$$

Then,

$$\begin{aligned} \text{(a)} \quad q \leq 0 &\Rightarrow 2(E[g(x)h(x)] - E(g(x))E(h(x))) \leq 0 \\ &\Rightarrow E(g(x)h(x)) \leq E(g(x))E(h(x)) \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad q \geq 0 &\Rightarrow E(g(x)h(x)) \geq E(g(x))E(h(x)) \end{aligned}$$